

Transfer Learning for Classifying Motor Imagery EEG: A Comparative Study

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Abstract—Motor imagery (MI) is the cognitive process when a person imagines performing a specific movement of their body. The corresponding electroencephalographic (EEG) signals can be measured externally to the head using scalp electrodes. Such signals have been applied in healthcare brain-computer-interface (BCI) systems, for example, motor rehabilitation and prosthetics control. These systems convert different MI EEG input signals to directives, so their performances depend on the efficiency of the embedded signal classification algorithm. In this paper we investigate the effectiveness of transfer learning in classifying the MI EEG data. The Continuous Wavelet Transform (CWT) is used to construct the scalograms, which serve as the inputs to the deep learning structure. The efficacies of five pre-trained networks—AlexNet, ResNet18, ResNet50, InceptionV3 and ShuffleNet—are evaluated on the BCI competition IV data set 2a. Binary (left hand vs. right hand) and four-class (left hand, right hand, both feet, and tongue) classifications are trained and tested using five-fold cross validation. The result indicates that using the CWT with transfer learning models provides very high classification accuracies. The ResNet18 network achieves the best accuracies in both cases at $95.03 \pm 2.95\%$ and $91.86 \pm 2.90\%$, respectively. In addition, we examine the effect of different time-frequency features on the classification performance by comparing the scalogram of the CWT and the spectrogram of the Short-Time Fourier Transform (STFT) as the inputs. It is found that the CWT is the preferred choice as it is superior to the STFT.

Index Terms—continuous wavelet transform, deep learning, electroencephalogram, motor imagery, scalogram.

I. INTRODUCTION

Neurocortical activities give rise to electrical signals that can be measured externally to the head called electroencephalography (EEG). One type of EEG recordings that is widely used is motor imagery (MI). The MI EEG is generated from spontaneous cortical activities without external stimulus while the subject imagines performing a specific movement without actually activating the muscles. The increase and decrease of EEG powers during the MI tasks are known as an Event-Related Synchronization (ERS) and Desynchronization (ERD), respectively. It has been shown that the ERD and ERS exhibit

different spatiotemporal patterns for different imaginary movements. [1].

MI EEG has been applied in many brain-computer-interface (BCI) systems due to its non invasiveness, excellent temporal resolution, low-cost hardware, and high portability. Examples of useful BCI systems in healthcare include automatic prosthetics or robot controls for transradial amputees [2], wheelchairs control [3], or motor rehabilitation for stroke patients [4]. Nonetheless, the MI EEG also offer some challenges, e.g., high dimensionality, non stationarity and low signal-to-noise ratio, that need to be tackled before it can be fully exploited.

The typical process for EEG-based BCI controls using MI starts with signal acquisition and pre-processing to remove noises and other unwanted artifacts. Next, the hidden features are extracted. Various techniques have been proposed for extracting features of the MI EEG signals. Spectral techniques such as Fourier transform isolate frequency information from the signal whereas time-based methods, for instance, autoregression analysis, extract the temporal information. Time-frequency approaches like Short-Time Fourier transform (STFT) and wavelet transforms obtain both temporal and spectral information simultaneously. Common spatial pattern focuses on the spatial information of the signal. Subsequently, the selected features form the input to the final classification stage, where its outputs are converted to hardware commands.

In previous decades, deep learning has emerged as a good candidate for MI EEG processing [5]. It is a family of machine learning algorithms comprising of multiple layers of neural networks, each attempting to mimic certain functions of the brain. Numerous studies have demonstrated that, with appropriate inputs and adequate amount of training data, the classification performance of deep learning algorithms can surpass classical techniques such as support vector machine, k-nearest neighbor, linear discriminant analysis, and artificial neural network. One popular class of deep learning methods

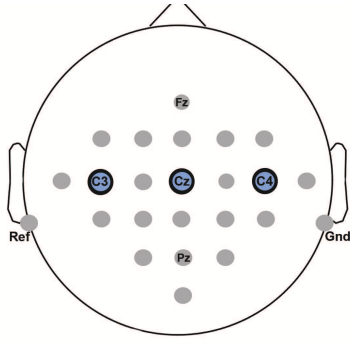


Fig. 1. Locations of the C3, Cz and C4 Electrodes.

is transfer learning. Instead of defining and training a new network structure from scratch, the method utilizes a network that has been previously trained on a large number of input data. The network then learns the new task by fine-tuning the network parameters. Hence, it can be trained faster and requires relatively small numbers of training data set. This is suitable for the BCI application with MI EEG since it can overcome the issue of a limited number of samples and reduce computation cost.

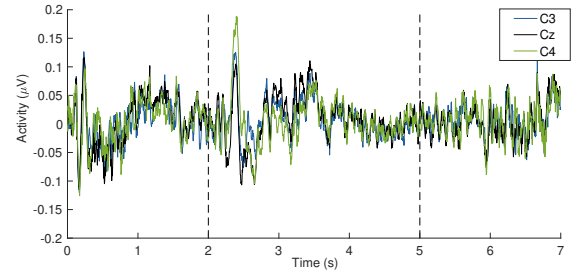
Although deep learning has the capability of fusing feature extraction and classification steps within its network structure, it is still a common exercise to extract distinctive features from the MI EEG signals and use those as inputs for the deep learning network. From our literature review, we discover that the use of Continuous Wavelet Transform (CWT) for feature extraction with transfer learning for classification has not been extensively studied. Hence, in this paper we examine the effectiveness of combining CWT and transfer learning for classifying the MI EEG data. We evaluate the efficacies of five pre-trained models for binary and four-class classifications. In addition, we investigate the effect of different time-frequency features on the classification performance by comparing the scalograms of the CWT and the spectrograms of the STFT as inputs for three selected transfer learning models.

II. METHODS

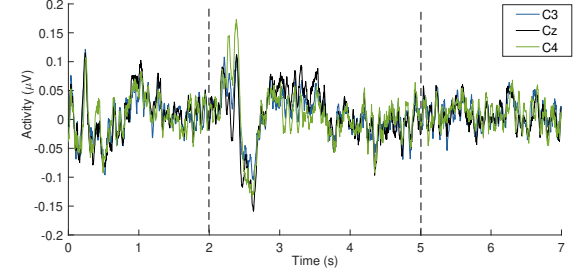
We are using Matlab as the main computation platform for both feature extraction and deep learning framework. All computations are performed using Nvidia Geforce RTX 2060 GPU on a notebook computer with Intel Core i7 CPU and 16GB of Ram.

A. MI EEG Data Set

We analyze the BCI Competition IV data set 2a in this paper. It is publicly available at <http://www.bbc.de/competition/iv/> and is one of the most used MI data set in literature. The data set contains 22-channel EEG recordings from nine healthy subjects. The sampling rate is 250 Hz. The data are bandpass filtered from 0.5-100 Hz and notch filtered at 50 Hz. The subjects perform four different imagined movements (left hand, right hand, both feet, and tongue) according to the on-screen visual cues. There are 144 trials in each class.



(a)



(b)

Fig. 2. Examples of trial-averaged data from Subject 9 imagining the movements of a) left hand and b) right hand. Only data from the three selected channels, C3, Cz, and C4, are shown here.

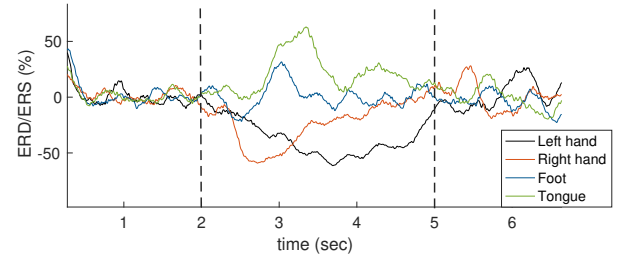


Fig. 3. Examples of the calculated ERD/ERS from subject 9.

Experimental paradigm and additional details of this data set can be found in [6].

We select the recordings from three electrodes— C3, Cz, and C4—to be processed. This also serves as an initial dimensionality reduction. As shown in the electrode montage in Fig. 1, the selected electrodes are over the primary motor cortex. Examples of trial-averaged data from Subject 9 imagining the movements of the left hand and right hand are depicted in Fig. 2. Figure 3 illustrates the ERD/ERS ratio. It is observed that the synchronization/desynchronization starts immediately after the visual cue onset at 2 seconds following the start of the trial, and the patterns for different movements vary significantly.

B. Feature Extraction using Continuous Wavelet Transform

To extract both temporal and spectral information of the signals simultaneously, we utilize the continuous wavelet transform [7] to construct the feature image. The concept of CWT is similar to the STFT. Instead of sinusoidal bases, it decomposes the signal over dilated and translated mother

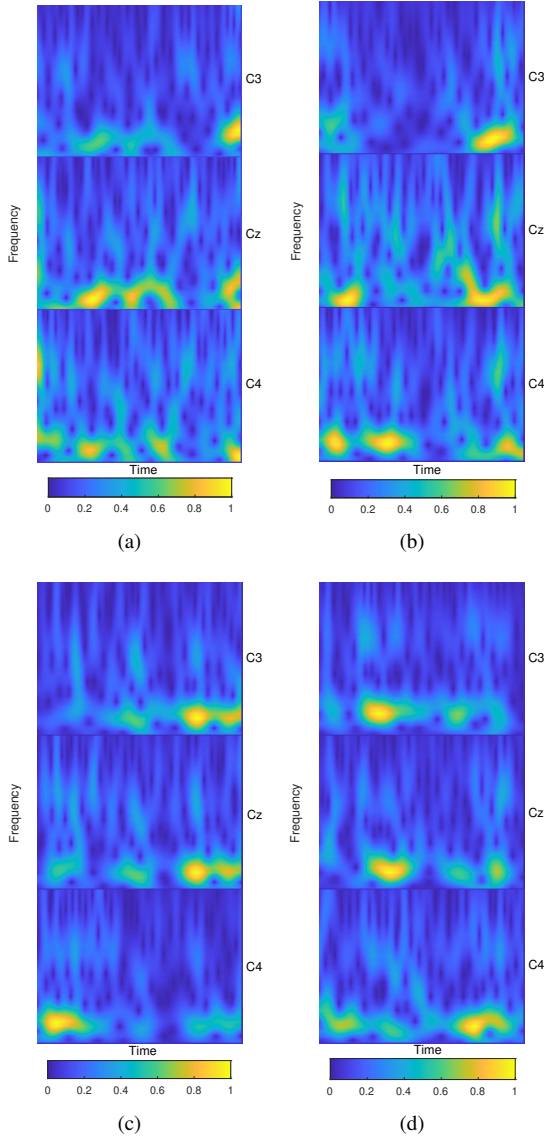


Fig. 4. Single-trial feature images of the four imaginary movements from Subject 9. (a) Left hand. (b) Right hand. (c) Both feet. (d) Tongue.

wavelets. Therefore, it has multi-resolution properties for both time and frequency scales and is suitable for dealing with non stationarity of the MI EEG signals.

Let $x_j(t)$ denotes the 1-D MI EEG recording from a single electrode for the j^{th} experimental trial. We compute the CWT coefficients $W(\tau, s)_j$ by projecting the signal within the 2 to 5 seconds time interval onto the wavelet atoms at different time and frequency resolutions using Eq. 1.

$$W(\tau, s)_j = \frac{1}{\sqrt{s}} \int x_j(t) \psi * \left(\frac{t - \tau}{s} \right) dt \quad (1)$$

where ψ is the analytic Morse mother wavelet [8]. The variables $s > 0$ and $\tau \in \mathcal{R}$ denote the scaling parameter and time-shift parameter, respectively.

It is widely known that the most prominent ERD/ERS patterns for MI are elicited within the μ -band (8-13 Hz) and

TABLE I
NUMBER OF NETWORK LAYERS AND COMPUTATION TIMES FOR
DIFFERENT PRE-TRAINED NETWORKS

	AlexNet	ResNet 18	ShuffleNet	ResNet 50	Inception V3
Number of Layers	8	18	50	50	48
Computation time (seconds)					
Two-class Training	71	70	106	137	221
Four-class Training	122	140	210	286	505
Two-class Testing	0.018	0.027	0.035	0.043	0.045
Four-class Testing	0.018	0.027	0.038	0.042	0.045

β -band (13-30 Hz). Hence, we truncate the results so that only the wavelet coefficients $W(\tau, s)$ in the 8-30 Hz frequency range are kept. The square of the magnitudes $|W(\tau, s)|^2$ are calculated and plotted, resulting in a scalogram of size 660×879 . The horizontal axis represents the duration from 2 to 5 seconds, whereas the vertical axis represents the 8-30 frequency range. The calculations are repeated for the recordings from C3, Cz, and C4 channels and the resulting scalograms are concatenated vertically to form the final MI EEG feature image of size 1980×879 . Exemplary feature images of the four imaginary movements are depicted in Fig. 4. It is evident that the four feature images have different patterns that can be used to distinguish between different imagined tasks.

C. Transfer Learning Models for Feature Classification

Five pre-trained Convolutional Neural Network (CNN) structures, namely AlexNet [9], ResNet18, ResNet50 [10], InceptionV3 [11] and ShuffleNet [12] are selected for our comparative study. These models have relatively small numbers of parameters so they can be optimized in a timely manner. AlexNet has the smallest structure among them with eight layers. It utilizes rectified linear units to achieve faster learning. ResNet18 and ResNet50 embody the residual networks with different numbers of CNN layers. The deeper ResNet50 also employs the bottleneck design. InceptionV3 attempts to avoid bottlenecks and utilizes grid size reduction as well as auxiliary classifiers. ShuffleNet introduces pointwise group convolutions and channel shuffle operation to regulate the flow of the data. The numbers of network layers are summarized in the top row of Table I.

To use the pre-trained networks for MI EEG classification, the feature image is reshaped so that it matches the required input dimension of each network. The fully connected layer and classification layer are modified to produce the correct number of classification outputs. During the training process, Adam is used as the optimization algorithm.

TABLE II

TWO-CLASS CLASSIFICATION ACCURACIES FOR DIFFERENT PRE-TRAINED CNN NETWORKS

Subject	Classification accuracy (%)				
	AlexNet	ResNet 50	Inception V3	ResNet 18	ShuffleNet
1	89.20	89.94	93.73	94.43	92.00
2	84.65	88.19	90.24	93.73	88.51
3	94.80	96.88	95.47	98.25	96.53
4	88.55	90.59	87.51	92.36	89.58
5	89.96	87.51	92.34	88.56	90.96
6	90.63	90.99	90.65	95.50	95.15
7	88.19	90.62	89.23	98.69	93.74
8	87.85	93.74	90.99	94.10	90.63
9	97.56	98.61	98.61	99.66	98.61
Mean	90.15	91.90	92.09	95.03	92.86
$\pm$ Std	± 5.35	± 4.23	± 4.21	± 2.95	± 4.16

III. RESULTS

A. Classification Accuracies of Different Pre-trained Models

We use the five pre-trained networks described in Section II-C to classify the MI EEG feature images and compare their classification accuracies. Five-fold cross validation is used to train and test each model. Specifically, each class of data is divided so that 80% of the input images are used to fine tune the model parameters. Ten percent of the images are used to validate the training results, and the remaining 10 percent is used to test the model for classification accuracy. We repeat this process five times and compute the overall classification accuracies.

For binary classification we classify between the left hand and right hand tasks. For multi-class classification we use all four classes: left hand, right hand, both feet, and tongue. The two-class and four-class classification accuracies for different pre-trained networks are reported in Table II and III. The ResNet18 network achieves the best accuracies for both cases at 95.03 ± 2.95 and 91.86 ± 2.90 , respectively. The performance of all networks degrades slightly when the complexity of the problem increases from two classes to four. As expected, the performance degradation is more pronounced for the networks with simpler structure, i.e. AlexNet and ResNet18.

The computation times for training the network and testing a single trial of data are presented in Table I along with the number of network layers within their architectures. It is observed that the training times for four-class classification approximately double those of the two-class cases. As expected, AlexNet and ResNet18, which have smaller numbers of layers, require less training times. ShuffleNet, ResNet50, and InceptionV3 have about the same number of network layers. Among these three models, InceptionV3 requires the most training time. The computation times for testing single-trial data between two-task and four-task classification are comparable.

B. Performance Comparison between CWT and STFT features

AlexNet, ResNet50 and InceptionV3 have been used in [13] with the same BCIC IV data set and similar pre-processing.

TABLE III

FOUR-CLASS CLASSIFICATION ACCURACIES FOR DIFFERENT PRE-TRAINED CNN NETWORKS

Subject	Classification accuracy (%)				
	AlexNet	ResNet 50	Inception V3	ResNet 18	ShuffleNet
1	84.39	92.01	91.14	90.98	89.41
2	82.12	90.80	89.23	90.97	90.80
3	90.62	92.70	91.84	93.92	93.22
4	82.82	90.80	86.64	92.01	89.23
5	78.82	89.94	86.99	90.29	89.24
6	82.82	91.49	86.64	92.01	90.98
7	82.98	92.88	89.77	92.53	92.19
8	81.25	89.59	84.21	89.76	88.55
9	90.80	94.27	94.79	94.27	95.48
Mean	84.07	91.61	89.03	91.86	91.01
$\pm$ Std	± 3.71	± 2.71	± 3.47	± 2.90	± 2.61

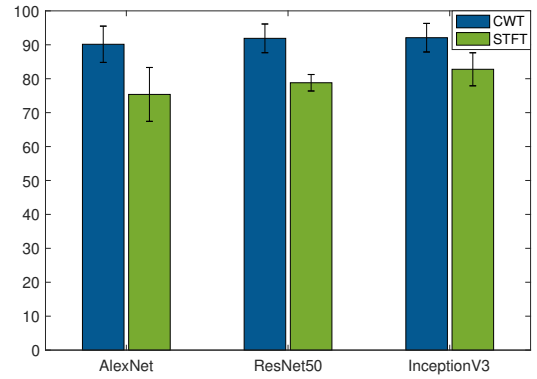


Fig. 5. Comparison between binary classification results (left hand vs. right hand) when using CWT vs. STFT for feature extraction.

However, they use short-time Fourier transform for feature extraction and form a feature image using stacked spectrograms. We compare their results of the binary classification between left hand and right hand movements with ours so that we can investigate the effect of different input features on the classification performance. The mean $\pm$ std classification results are depicted in Fig. 5. It is noticed that, across all models, the use of the CWT provides superior classification results than when the STFT is used. This is reasonable due to the fact that although the STFT is also a time-frequency representation, it suffers from the trade-off between the temporal and frequency resolutions because the window size is fixed. On the other hand, the CWT varies both frequency scales and time scales to capture the information within the signal.

IV. DISCUSSION AND CONCLUSION

In this paper we have investigated the efficacy of using transfer learning to classify MI EEG data by comparing performances of five pre-trained networks. The scalograms of the CWT serve as the input images of the deep learning models. It is evident that using the CWT with pre-trained network is effective. It attains excellent accuracies for both binary and multi-class classifications. It is interesting to note

that the network that performs the best, i.e. ResNet18, is not the deepest network. This may be because of the limited size of training data sets used for training the networks: architectures with more layers requires more data to properly optimize the network parameters. This finding is advantageous for the BCI system because employing a smaller network is more computationally efficient. Thus, the system will have better real-time response. Lower computational cost also prolongs battery life of the hardware.

As expected, the choice of feature extraction technique affects the classification performance. It is found that the CWT is the preferred choice as it is superior to the STFT. More extensive investigation is needed to find the MI EEG features that work best with transfer learning to further improve the BCI systems.

REFERENCES

- [1] G. Pfurtscheller, C. Brunner, A. Schlögl, and F. Lopes da Silva, "Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks," *NeuroImage*, vol. 31, no. 1, pp. 153–159, 2006. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1053811905025140>
- [2] J. Zhang and M. Wang, "A survey on robots controlled by motor imagery brain-computer interfaces," *Cognitive Robotics*, vol. 1, pp. 12–24, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S266724132100001X>
- [3] K. Choi and A. Cichocki, "Control of a wheelchair by motor imagery in real time," in *Intelligent Data Engineering and Automated Learning – IDEAL 2008*, C. Fyfe, D. Kim, S.-Y. Lee, and H. Yin, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 330–337.
- [4] M. Sebastián-Romagosa, W. Cho, R. Ortner, N. Murovec, T. Von Oertzen, K. Kamada, B. Z. Allison, and C. Guger, "Brain computer interface treatment for motor rehabilitation of upper extremity of stroke patients—a feasibility study," *Frontiers in Neuroscience*, vol. 14, p. 1056, 2020. [Online]. Available: <https://www.frontiersin.org/article/10.3389/fnins.2020.591435>
- [5] A. Al-Saegh, S. A. Dawwd, and J. M. Abdul-Jabbar, "Deep learning for motor imagery EEG-based classification: A review," *Biomedical Signal Processing and Control*, vol. 63, p. 102172, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1746809420303116>
- [6] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. Müller, G. Mueller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, and B. Blankertz, "Review of the BCI Competition IV," *Frontiers in Neuroscience*, vol. 6, p. 55, 2012. [Online]. Available: <https://www.frontiersin.org/article/10.3389/fnins.2012.00055>
- [7] S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*. Academic Press, 2008.
- [8] S. Olhede and A. Walden, "Generalized morse wavelets," *IEEE Transactions on Signal Processing*, vol. 50, no. 11, pp. 2661–2670, 2002.
- [9] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Communications of the ACM*, vol. 60, pp. 84 – 90, 2012.
- [10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [11] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the Inception architecture for computer vision," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Los Alamitos, CA, USA: IEEE Computer Society, Jun 2016, pp. 2818–2826. [Online]. Available: <https://doi.ieeecomputersociety.org/10.1109/CVPR.2016.308>
- [12] X. Zhang, X. Zhou, M. Lin, and J. Sun, "Shufflenet: An extremely efficient convolutional neural network for mobile devices," in *2018 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2018, pp. 6848–6856.
- [13] N. Shajil, M. Sasikala, and A. M. Arunnagiri, "Deep learning classification of two-class motor imagery EEG signals using transfer learning," in *2020 International Conference on e-Health and Bioengineering (EHB)*, 2020, pp. 1–4.